

2.4 Traditional Education System

The struggle for understanding the riddles of the universe and matter is continuous since the dawn of human civilization. By living in close proximity to nature for centuries, the indigenous people gathered an immense knowledge base from the environment in which they used to live. They developed a deep understanding of the properties of plants and animals surrounding them. Living in the variety of complex ecosystems, during generations to generations it became an important element of their cultural identity. The knowledge gained by them and passed on through traditions is the holistic one, and cannot be separated from the people who hold it. The knowledge is based on the spiritual health, social and cultural well-being of the people as a way of their life.

The traditional knowledge is the understanding gained by the local people at a particular region over a long period of continuous interaction with nature and environment. They developed an accrued knowledge system of beliefs and values, and imparted the wisdom to the generations through teachings and practices irrespective of its scientific validity. In the beginning, the knowledge was passed orally and practically from generation to generation through cultural and social practices i.e. folklores, rituals, stories, songs, ballads, daily-routine activities and sayings of the local people possessing it. Later on, the indigenous scripts were developed, and specific texts were written.

Ever since the time immemorial Bhārata has not only been the leader in religion, philosophy, and metaphysical knowledge, but also made seminal contributions in science and technology, trade, commerce, fine art, music, dance, theater, drama, health, sports and archery. Yoga, Āyurveda, martial arts (e.g. Kung Fu in China) and chess are the well known gifts of Bhārata to the world. The age-old thriving civilization of Bhārata has a unique place in the world; i.e. known as the 'Ved Bhūmi' – the land of knowledge. It has not only survived off the test of time, but also preserved the vast knowledge system over the centuries, and that too as the regular practices of common people. No doubt, almost all the civilizations have contributed to shape the evolution of mankind, i.e. moulding and remolding the individuals and societies through the long trail of progress and development, but unfortunately, the course of time has led most of the civilizations to extinction. However, Bhārata civilization is still standing on the same age-old foundation. The heritage of the knowledge system of Bhārata has been passed down through the thousands of years, and largely preserved by the inscriptions on the temples, manuscripts and the education centers i.e. 'gurūkulas'. This could survive amidst the ruthless course of numerous challenges through centuries mainly due to her vibrant and scientific basis of the traditional knowledge system.

Structure of Learning Centers and Specialties

In Bhārata, the concept of traditional knowledge and practices is quite old, and undoubtedly, the land has developed a vast traditional knowledge system in the different pockets of communities abiding in varied environments. The knowledge gained by the communities during the interaction with nature and environment must have been supplemented by the existing rich knowledge system. The knowledge

system so developed was not only systematic and scientific, but also holds true in all times and climes.

This knowledge has been put in practice and has formed an integral part of the daily life and routines, i.e. the rituals and beliefs of various communities of Bhārata from time to time. However, over a long journey in time, the very scientific foundation of this knowledge was forgotten, and the people of land carried these traditions by default without knowing of the scientific significance behind that.

It is this strength that made Ancient Bhārata highly advanced and to be called as the cradle of the world civilizations over the ages. After getting the enlightened knowledge of the Vedas, Bhārata became the epicentre of almost all the other sciences of the world. It has been recorded that during the Vedic period till the time of the Mahābhārata war, science and technology were at its advanced stage. This culture of education, science and technology was alive during the struggle period¹⁵ and remained vibrant even till the beginning of conflict with the Britishers and other Europeans.

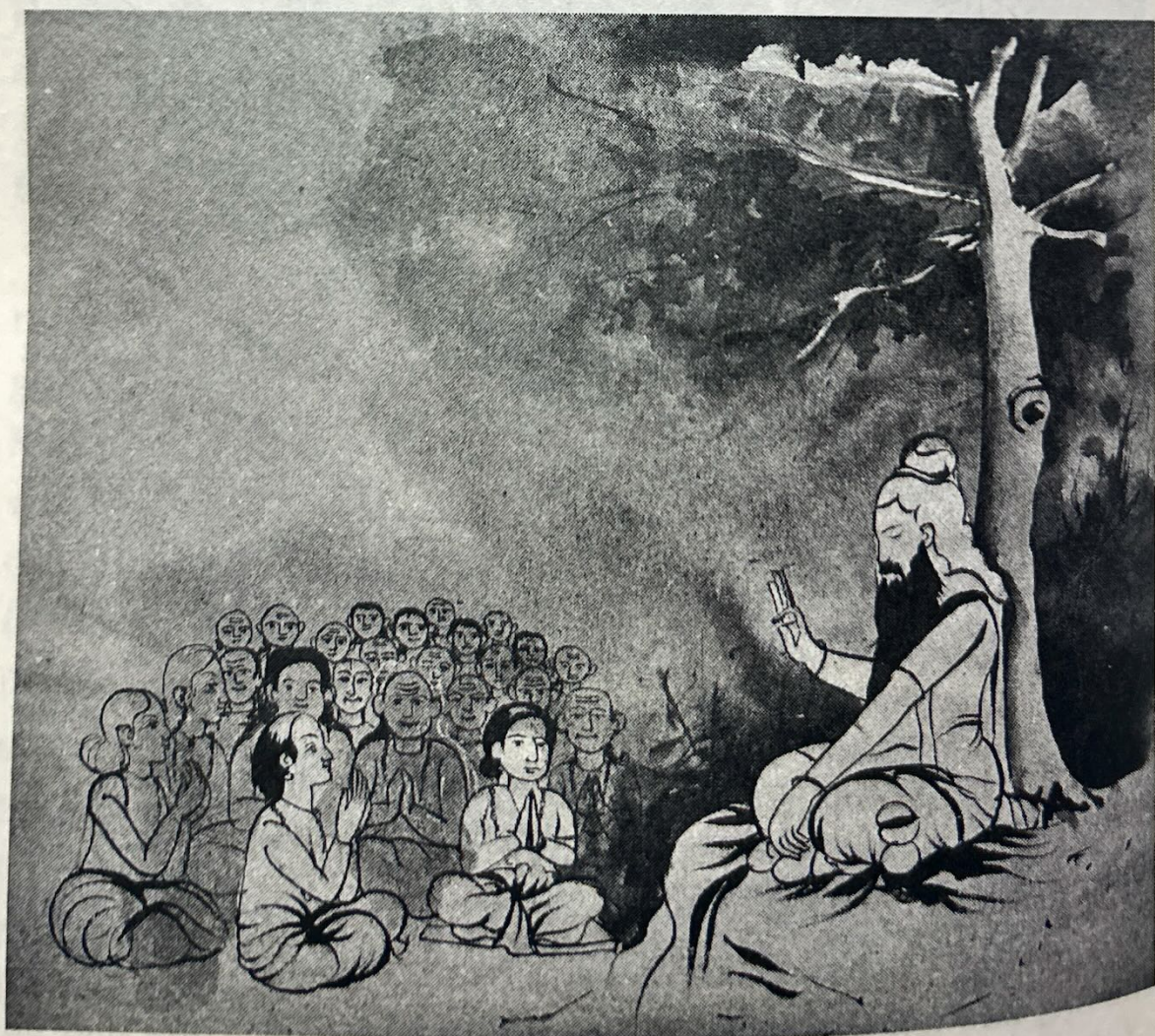
Bhārata was the nucleus of education for the entire world, as there were several universities, learning centers, and schools - the gurūkulas. According to the survey on indigenous education in Madras presidency (1822-25) conducted by Sir Thomas Munro the state of education was much higher than most of the European countries. There was hardly a single village without a school. Similar facts were observed in the survey reports of other parts of Bhārata like Maharashtra (1835), Bengal, Punjab, and Bihar (1838). The status of education in Bengal was conducted during 1830-40 CE and the first report of William Adam came up in 1835 and the 2nd and 3rd in the years 1838 CE. In 1790 CE, Navadveep University also conducted a survey, according to which there were 150 teachers for 1100 students.



The knowledge system that discusses the past helps us to live our present in a better way, and may further guide us to future. Some may argue against it, but there is a proven fact that a civilization can neither sustain in present without past, nor grow or move into future with our empty hands. Bhārata has a rich tradition of information gathering and knowledge acquiring, and finally converting that all

into wisdom. As stated earlier, the intellectual inquiry and the textual heritage of this land dates back to thousands of years.

As such, Bhārata was highly advanced in knowledge traditions and practices since the ancient times. In fact, scientific tradition continued even during the struggling phase of Bhārata with invaders and the colonial periods. In the early colonial phase, for their interests the British rulers needed surveys of geology, botany, zoology, and meteorology and they trained the Bhāratīya just to assist them. However, at the turn of 20th century, they changed the policy of science and technology for higher education and that, in fact, accelerated the spread of modern science in Bhārata. For more details, one can see Dharampal's works and proceedings of the conference held at the Indian National Science Academy (INSA), New Delhi, during 14 – 16 March 2018¹⁶.



The educational awareness in the society, i.e. fervor about the knowledge acquiring can be seen from the fact that the education for all was not a state subject, but it was a public concern in deed. To promote education amongst all the individuals i.e. the education for common people, knowledge of elementary Bhāratīya philosophy, literature, and elementary mathematics was the popular subjects and the jobs were entrusted to skilled and subject-specific teachers residing in the big villages or temple towns. The donations, property owners, local village headmen, merchants supported these gurūkulas; however, some part of the expenses was met from student fees. To learn and disseminate the knowledge was a matter of great pride. To impart the basic education for all i.e. knowledge of Bhāratīya philosophy, literature, and elementary mathematics, a class of good teachers was appointed at the village and town level with the social support. This shows that the Bhāratīya valued knowledge and knowledge system above anything else.

According to Suhas Chatterjee¹⁷, the 'chatūrāśrama' was the dominant social structure that prevailed in Bhārata since the Vedic times to the subsequent periods. In this system, the disciples from the age of six to twenty-four used to receive instructions by sitting in front of and close to the feet of the preceptors in the hermitage. Soon after, it was felt that one or two teachers in the hermitage were not in a position to teach all the subjects properly. As such, they realized the need of a bigger centre of learning in a well-developed and organized way. Therefore, they planned higher learning centers in the villages and temple towns, and the bigger and biggest monasteries to teach a large number of students by a number of preceptors with a variety of subjects. In the ancient times, the nomenclature for higher education institution (university in modern days) was 'Maṭha' i.e. the monastery.

The great universities in Bhārata¹³, e.g. the Takṣaśilā and Nālandā, were open for the entire world. However, there were many other

learning centers comparatively of lower standard. The specialties of these education institutions were as:

- Vikramśilā near Bhāgalpur (specialized in logic, grammar and metaphysics),
- Valabhī in Gujarat (philosophy, language, science and economics were main subjects),
- Ujjain (centre of mathematics and astronomy, etc...),
- Vārāṇasī (authority of Saṃskṛta, religion and philosophy),

There were also other centers, like Odāntapurī and Pātliputra in Bihar, and Rājgīra, Bodhgayā, Jālandhara, Avanti, Mathurā, Kāncī, and Kannauja etc... They used to teach the Vedas and its attendant philosophy and grammar. The other important branches included in the curriculum were physical sciences, medical sciences, mathematics, astronomy, engineering, architecture, political and economic history, literature, philosophy, and religion. Religion in Bhārata was not just theology; it rather included subjects like grammar, logic, and astronomy.

It has been noted that under the exigencies of large-scale battles during the epic period, the subjects like physical education and medical sciences developed considerably. Despite of the secular nature of sciences, the monasteries were under the complete control of monks, and hence the studies of Mahāyān School of Buddhism and Hinduism, and spiritual studies receive attention as the greatest branch of human knowledge.

Admission

The gurūkulas were established to educate the common masses to the royal class families. The meritorious students of the known world used to crave for admission into the prestigious institutions of

Bhārata. The religion, class and sex were not factors in seeking admission to the educational institutions.

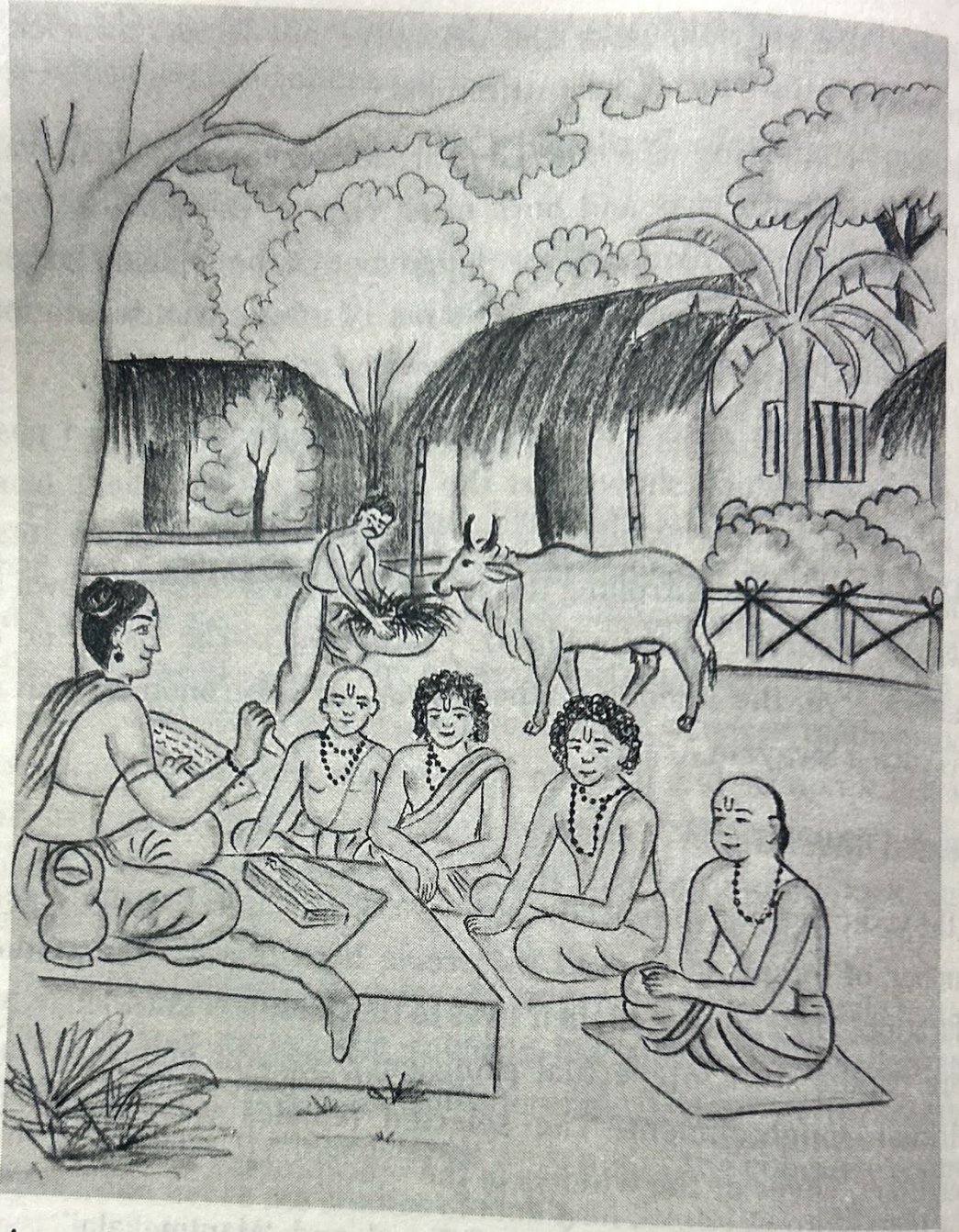
After conferring the Sacred Thread ceremony, 'upanayana saṃkāra' on both girls and boys of all classes, the children were admitted to the gurūkulas. After the advent of the female teachers, called 'Upādhyāyanī', increasing number of educational institutions were dedicated to girls pursuing higher education.

Although rare, there are instances during the Vedic and post-Vedic period, which shows that the students with mother's name alone (father's name wanting) were admitted to school. Some noted females were also controlling the schools, which were also known as 'Swayam Ācārya' (upādhyāyanī/ preceptress). The daughter of Bhāskrācārya, the famous mathematician and astronomer, Līlāvati was such a Swayam Ācārya.

Women Education

In the vedic period women education was given equal priority. A good number of young girls used to receive both literary and cultural education. They dedicated their lives to the pursuit of knowledge, and were paragons of intellectual proficiency, natural philosophy and spiritual enlightenment. The selected females were either the daughters, wives or the widows of the preceptors.

In Tamil literature, 'Silappatikāram' and 'Maṇimekalai' is the world's only epics with women as the protagonists. The Vedic literature 'Mahānirvāṇa Tantra' admires the birth of a scholarly daughter the a family as: A girl also should be brought up and educated with great effort and care. According to Devī Mahātmya all forms of knowledge are aspects of Thee; and all women throughout the world are Thy forms.



A good number of women-ṛṣis; i.e. viduṣīs contributed to the development of the literature. In the Ṛgveda about thirty (30) female-seers (viduṣīs) have contributed substantially. These viduṣīs were also called as 'brahmavādinīs', i.e. women ascetics; those who strive for understanding the highest philosophical knowledge- Super-Consciousness. There are some names, who contributed largely to the Vedic ethos: Ghoṣā, Lopāmudrā, Maitreyī, Gārgī Kakṣivati, Dakṣiṇā Prajapathya, Viśvāvar, Atreyī, Apālā, Kādrū, Yami, Romaśā, Aditi,

Dakṣāyaṇī, Juhū, Rātri Bhāradwājā, Sāvitri, Indrāṇi, Nodhā, Urvaśī, Paulomī, Śāśwatī, Aṅgirasi, Śrī Lakṣa, and Līlāvatī.

Some Common Features of Education in the Universities

- The external authorities like kings or local leaders had no control over the scholastic activities.
- The teachers had intellectual freedom; i.e. complete autonomy in work, tutoring any number of students as per choice and teaching the subjects of preference without conforming to any centralized syllabus.
- Teacher's satisfaction depended on the achievement level of students to complete the program of study.
- The students, who were unable to fit into the social, intellectual, and moral atmosphere of the gurūkula, were advised by the teacher to quit their studies.
- It was only the satisfaction of the teacher and there were no conditions of the examinations for students, to complete their studies.
- The student was allowed to proceed to the next level, once the thorough mastery of previous unit was achieved.

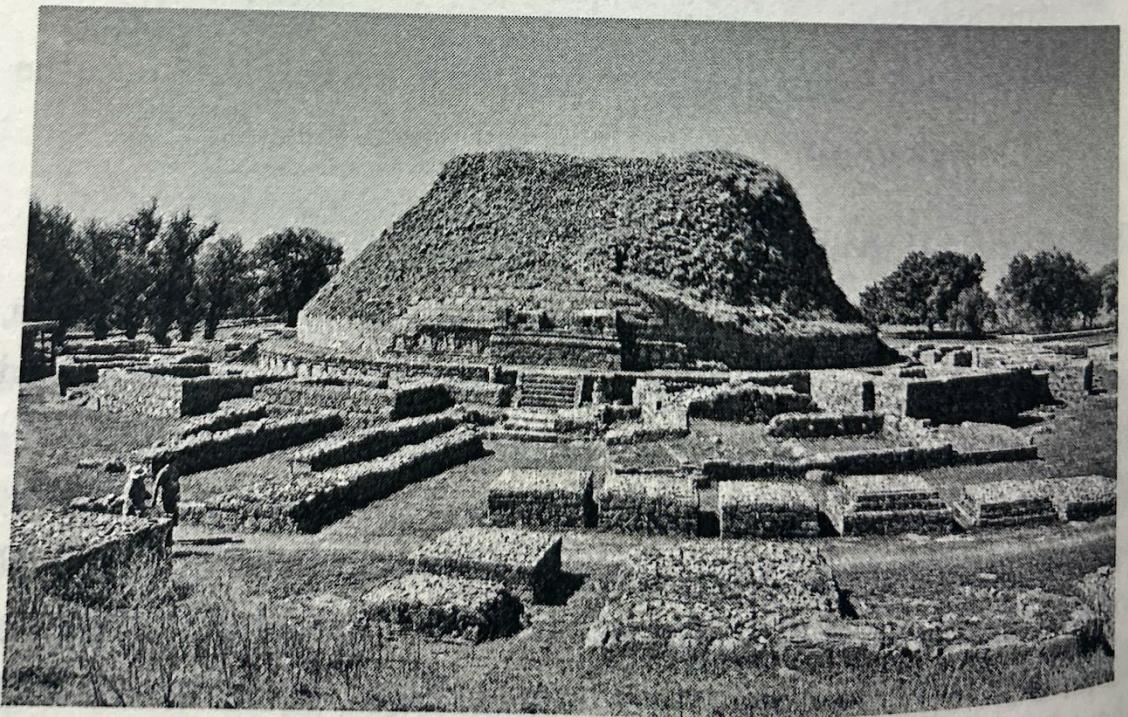
The Takṣaśilā University

The knowledge being precious and sacred to Ancient Bhārata, therefore it is not surprising that the world's first university named the 'Takṣaśilā University' was established in Bhārata around 1000 - 700 BCE. Takṣaśilā was the capital of Gāndhār state situated in the western part of Ancient Bhārata, which now falls in Pakistan. However, some historians date the existence of this university back to the 6th - 7th century BCE. This was a residential centre of learning housing about 7000 students and teachers from the larger part of the world, including Babylon, Greek, Arabs, Iraq, Iran, Syria, and China. During

the olden days, the Takṣaśilā was equivalent to the MIT, IIT kind of institutes. The university was world famous for studies in science and medicine.

Subjects Taught in the University

During the Maurya reign, the Takṣaśilā University continued as a great centre of disseminating knowledge to the deserving students. In the university the students used to learn subjects including Vedas, Āyurveda, agriculture, surgery, polity, archery, warfare, astronomy, science, mathematics, commerce, futurology, grammar, philosophy, music, and dance etc. There were also some snoopng subjects like the art of discovering hidden treasure and decrypting encrypted messages, etc. There were several seminar halls, prayer halls and classrooms, well-equipped laboratories for physical sciences and medical sciences, sophisticated operation theaters, up to date observatories for studies of astronomical objects and solar system, workshops for engineering and architectures and a big library containing thousands of manuscripts.



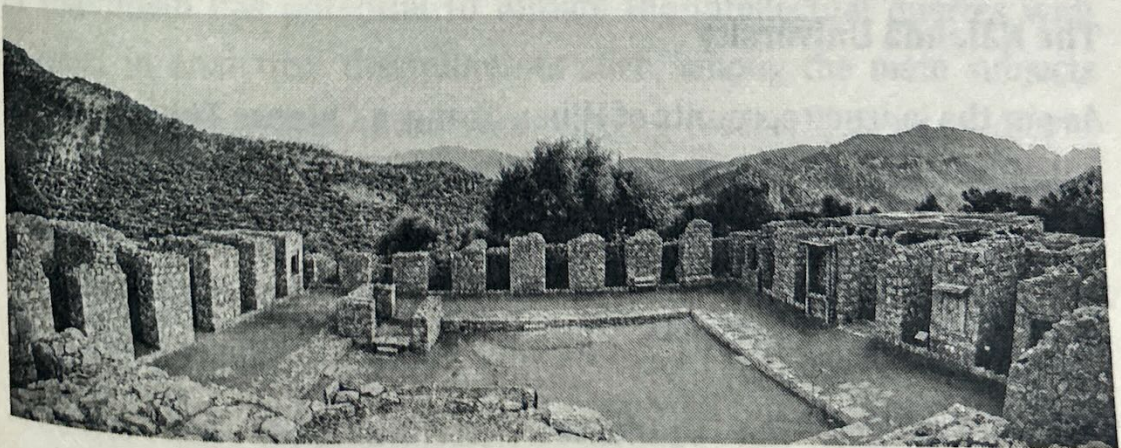
The ruins of Takṣaśilā university [Courtesy: Commons.wikimedia.org]

Alumni of the University

The Emperor Kaniṣka and the host of scholars like Nāgārjuna and Aśvaghoṣa, etc..., was closely associated with the university. Famous teachers and pupils, like Pāṇini, Kauṭīlya, Prasenaṇita, Viṣṇu Sharma, Jīvaka, and Chandragupta Maurya etc... were the among the great alumni of the university:

- Cāṇakya (Kauṭīlya/ Viṣṇugupta), who wrote the world's finest work till today on political duties, statecrafts, economic policies, state intelligence system, administrative skills and military strategy, called 'Arthaśāstra';
- Pāṇini, the famous Saṃskṛta grammarian, who created a scientific treatise on language and grammar - Aṣṭādhyāyī;
- Viṣṇu Sharma, who authored the collection short moral stories as 'Pañca-Tantra';
- Jīvaka, who performed here, for the first time in the world, the brain surgery;
- Nāgārjuna also conceived of in a crude form the theory of relativity, which revolutionized the modern physics under the formulation of Albert Einstein;
- Chandragupta Maurya was the ruler.

During the course of time for centuries, the students of Takṣaśilā became famous and top brands.



The ruins of Takṣaśilā university [Courtesy: Commons.wikimedia.org]

The Last Phase of the University

Possibly, after Huns invasion, the atmosphere of Takṣaśilā was full of turmoil and the university, which was a brand name for specialization in science, medicine, and architecture, now ceased to be so. Certainly, it was unable to offer a peaceful and congenial environment to the scholars for excelling in science and research, and practicing spirituality and other subjects. Therefore, a need for other similar centers of prominence cropped up. Like Nālandā, Ujjain became the centre for mathematics and astronomy. Vārāṇasī continued to attract the scholars of Saṃskṛta, grammar, religion and philosophy.

Establishment of the Alexandria University

The records of Greek writers show that when Alexander came to Bhārata he was highly impressed by the system of education in the Takṣaśilā University. He had a Bhāratiya physician in his camp to treat the wounded and ailing of the Greek soldiers. Being inspired by that, he was curious to establish this type of learning centre i.e. a great university at Alexandria in Egypt. As such, the famous Alexandria University was established in 300 BCE. In the first university ever built outside of Bhārata, scholars from Greek, Iran, Bhārata, and Egypt used to study and teach. It is held that a large number of Bhāratiya texts were translated into Greek for to be kept in the library of Alexandria.

The Nālandā University

As per the journey accounts of Hiuen Tsang, a Chinese Traveller¹⁸, the Nālandā University was another big learning centre in Ancient Bhārata established during the early 500 BCE in Bihar between Gaya and Patna. Until the violent destruction by the ruthless looter Bakhtiyar Khilji in 1193 CE, the university was one of the largest universities in Asia and Europe.

Education and Infrastructures of the University

The world's first six storied residential university used to accommodate about 10,500 students with big hostels. In the university, called as the Mahāvihāra, there were 100 classrooms and about 1000 professors taught more than 60 subjects.



The ruins of Nālandā university [Courtesy: Commons.wikimedia.org]

As a Buddhist foundation, this created much space for the pursuit of analytical and scientific subjects within the campus. The philosophy of Buddhism has potential to initiate the intellectual queries with interest in analytical discipline. As such, among the main subjects taught, and on which there was ongoing research, were medicine, astronomy, public health, architecture, sculpture, religion, history, law, and linguistics.

There were also several seminar halls, prayer halls, well-equipped laboratories for physical sciences, medical sciences, operation theaters, and observatories, workshops for engineering and architectures, classrooms for polity, religion and philosophy. A large

library containing thousands of manuscripts kept in three big halls was called the 'mart of knowledge'. The Gupta rulers, who had made some significant amount of donations, supported this university and Harṣa donated the income of 100 villages for its maintenance.

Admission

In his journey account, Hiuen Tsang has also mentioned similar things as discussed above, and specifically mentioned about the cordial people, beautiful buildings, and high hanging observatories in Nālandā. In 700 CE, Nālandā was known as one of the largest and famous universities in the world; however, the European universities like Oxford, Cambridge, and Wittenberg came up during 11th -13th centuries. Admission was rather tough, as students had to pass through a test of very high standard. Just 20-30 % of the total candidates were qualifying those tests. A student obtaining degree from this university was a matter of great pride, and a surety for getting a prestigious job anywhere in the world. To earn good amount of money and respect, the students used to pretend to be the graduate of the Nālandā University.

Famous Alumnii of the University

- The Chinese students of the university, such as Xuanzang and Yi Jing, who were some of many translators of Saṃskṛta texts into Chinese, wrote that Nālandā was a Buddhist foundation, as were Vikramśilā and Odāntapurī, where the central focus were the studies of Buddhist philosophy and practice.
- Much is not known about the education of Āryabhaṭa, the great mathematician, and astronomer, but some accounts described him as head of the Nālandā University. However, after the fall of Mauryan Empire he might have taken refuge in the southwest coast of Bhārata.

- Asaṅga and Vasūbandhu were the famous professors of this university, both religious and secular subjects were taught to the qualified students, and the high standard of intellectual life was maintained in the university.

Seminars and Debates

The Chinese scholars describe occasions where the scholars from various corners of the world used to assemble for a week or more and take part in debates and discussions on specific topics. As said before, there were a number of seminar halls in these universities and other learning centers to accomplish such kind of jobs. These kinds of meetings were the real tests of the scholars' understanding and quick responsiveness on a particular issue. For the graduates of these universities, mere cramming was not enough to stand in debates against their opponents. Otherwise, they might not have impressed everyone for long.



The ruins of Nālandā university [Courtesy: Commons.wikimedia.org]

Exceptional Gems

Despite of the great centers of learning, there were eminent teachers and scholars in different kingdoms, towns and villages, who did not join the great institutions to impart education. They contributed a lot in the field of mathematics, science, astronomy, art, culture and polity and others fields of knowledge.

Amongst the eminent teachers were Patañjali, the famous grammarian and physician; Manu, the famous lawgiver; Caraka and Suśruta, the great physicians; and Dhanvantri, the legendary physician, Bhāskarācārya, Varāhamihira, Brahmagupta, and Mahāvīra etc...

The high status and honor of the big institutions like Takṣaśilā, Nālandā and other seminaries of monasteries order did not attract them. They were institutions by themselves, and had considerable followings.

Knowledge Export¹³

In the 1st millennium BCE, Iran was virtually an expansion of Bhāratiya culture and civilization. At the western fringe of it was Asia Minor (modern Turkey), which acted as a square of interaction between Greeks and Iranians. During the 6th century BCE, Iran expanded its western borders to include Assyria, Babylon, whole of Asia Minor, and the main part of Greek and Egypt soon after. At that time, the eastern part of Iran was in peace and continuously updating with the Bhāratiya knowledge and traditions.

The spread of Bhāratiya culture largely owed to the students and teachers of these educational institutions, through which Bhārata held that place of honor and leadership amongst the major part of the known world. Certainly, it was not the sword of military action and fanaticism to do that. Contributions of the Bhāratiya education acted as the torchbearers of knowledge, which broadened the horizon of

human awareness and thus enlightened the human mind for the centuries to come.

Apart from these, the wandering monks travelled across the length and breadth of the holy land. They extensively added to the expansion of Bhāratīya knowledge in Asia and Europe. Moreover, Bhāratīya sea-traders substantially contributed to the knowledge export from this end. In this way the religions, philosophies, and science travelled from Bhārata to the West in the ancient times. This had happened certainly neither by the sword and nor by bigotry and fanatics. As such, Bhārata could hold the highest place of honor and leadership in the yore.

In the chronicles of Greek and Chinese Travellers, it has been written that the Muslims from Central Asia believed that Bhārata was a paradise with its perfumes and medicines. They were also scared of the knowledge that the Bhāratīya had. They wanted to destroy what they thought was a superior civilization. They targeted the universities and places of learning, since education had brought prosperity to these kingdoms.

Finally, all the great learning centers Takṣaśilā, Nālandā, Vikramaśilā, Odāntapurī, and other Bhāratīya were brutally deserted and destroyed by the invaders. All the inmates were killed and the literature was burnt by the extremists and fanatics, and thus pushed Bhārata into the period of darkness. Similarly, most of the places of learning were destroyed; the Alexandria in the West too met the same fate at the hands of the enemies of civilization.. Despite the existing great institutions and other proliferated learning centers, the system of education under the individual preceptors did not die out altogether.

Around the 8th century CE, transfer of knowledge from Bhārata to Arab started and translation of Bhāratīya texts took place at a massive scale. The Arabs ruled over a vast area from Sindhu to Spain in the 8th

century, when there was continuous importing of knowledge from Bhārata. Incidentally, some of Bhāratiya works especially in mathematics, astronomy, astrology, medicine and philosophy, etc. had already been translated into Arabic. In this way, the Arabic version of Bhāratiya knowledge spread all over the world particularly to Europe. Many such texts were translated into Latin and other European languages.

Typical examples are works on the decimal system of numbers and their speedy computation attracted Fibonacci (1170-1250 CE), and 'Oupnekhat' (Book of the Secret), a Persian translation of Upaniṣads 'Sirr-e-Akbar' (The Greatest Mystery) by Dārā Śikoh deeply touched and inspired the great German philosopher Arthur Schopenhauer (1788-1860 CE). He called this work as the greatest gift of the century, and predicted the philosophy and knowledge of the Upaniṣads to become the cherished faith of the Westerners.

Review Questions

- 1) Discuss the genesis of the Bhāratabhūmi and Saraswatī-Sindhu Civilization.
- 2) The Saraswatī River is not a myth; justify this statement with the evidences, which support its real existence.
- 3) Write various reasons in support of the antiquity of Vedic culture.
- 4) List down the various universities existing in Ancient Bhārata.